Data-Driven Methods using Parameterized Robust Optimization

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Feb. 16, 2024

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Collaborators (for this research)

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Tianyu Dai (Mathworks)

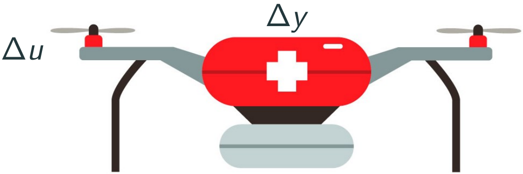
Motivating Scenario

Flying a drone

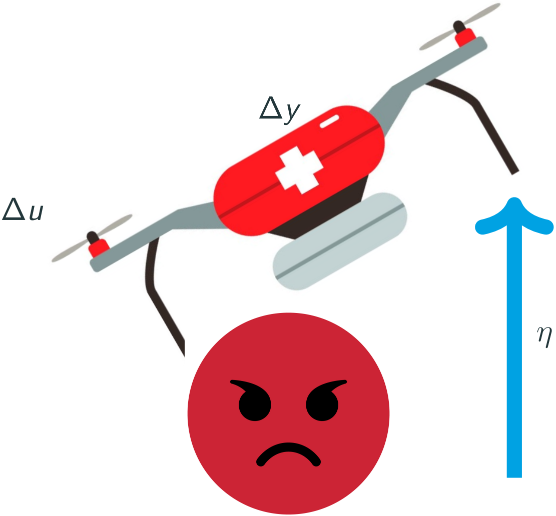
Flying a drone

Flying a drone

Flying a drone

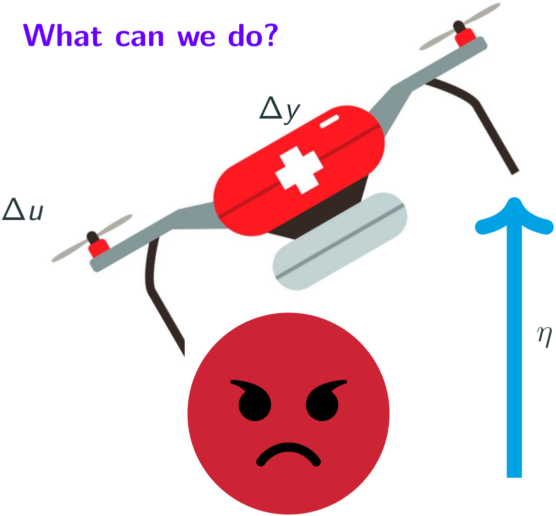


Flying a drone



Flying a drone

What can we do?



Errors-in-Variable Noise Task

Noisy measurements $\mathcal{D} = \{\hat{x}_t, \hat{u}_t\}_{t=1}^{T}$ of linear system ($y = x$)

$$
\begin{aligned}
x_{t+1} &= Ax_t + Bu_t + E\eta_t && \text{(1a)} \\
\hat{x}_t &= x_t + \Delta x_t && \text{(1b)} \\
\hat{u}_t &= u_t + \Delta u_t && \text{(1c)}
\end{aligned}
$$

Errors-in-Variables (EIV): $\Delta x, \Delta u \neq 0$

Want to control system under EIV noise

Bilinear Trouble

$(A, B, \Delta x, \Delta u, \eta)$ all unknown

$$\hat{x}_{t+1} - \Delta x_{t+1} = A\hat{x}_t - A\Delta x_t + Bu_t - B\Delta u_t - E\eta_t$$

Multiplication between unknown $A\Delta x_t$, also in $B\Delta u_t$

Nonconvex identification, control problems [1]

[1] T. Söderström. Errors-in-Variables Methods in System Identification. Springer, 2018

What can we do for EIV control?

Behavioral: (soft noise bounds)

DeepC: Regularize against errors-in-variables [2]

Set Membership: (hard noise bounds)

Prior: Ignore EIV, abstract noise into η

Ours: Address EIV directly, and pay the price[3]

[2] I. Markovsky, L. Huang, F. Dörfler. Data-driven control based on the behavioral approach: From theory to applications in power systems. IEEE Control Systems Magazine, 2023

[3] Jared Miller, Tianyu Dai, and Mario Sznaier. Data-Driven Superstabilizing Control of Error-in-Variables Discrete-Time Linear Systems. In IEEE CDC, 2022

Flow of Presentation

Present Set-Membership EIV stabilization method

Reduce complexity with parameterized robust optimization

Apply methods to OCPs and related problems

EIV Sets and Control Problem

Consistency Set

Consistency set $\bar{\mathcal{P}}(A, B, \Delta x, \Delta u)$ (with $\epsilon_\eta = 0$)[4]

$$\bar{\mathcal{P}} : \left\{ \begin{matrix} 0 = -\Delta x_{t+1} + A\Delta x_t + B\Delta u_t + h_t^0 & \forall t = 1..T-1 \\ \|\Delta x_t\|_\infty \leq \epsilon_x, \ \|\Delta u_t\|_\infty \leq \epsilon_u & \forall t = 1..T \end{matrix} \right\}$$

Affine weight h^0 (residual) is defined by

$$h_t^0 = \hat{x}_{t+1} - A\hat{x}_t - Bu_t \qquad \forall t = 1..T-1$$

[4] V. Cerone. Feasible parameter set for linear models with bounded errors in all variables. Automatica, 29(6): 1551–1555, 1993.

Stabilizing the Plants

Desired control policy $u = Kx$

Set of plants consistent with $\mathcal{D}$ (with projection π):

$$\mathcal{P}(A, B) = \pi^{A,B}\bar{\mathcal{P}}(A, B, \Delta x, \Delta u)$$

Control Problem:

Find $K \in \mathbb{R}^{m \times n}$ such that $(A + BK)$ is Schur $\forall (A, B) \in \mathcal{P}$

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Schur stability (with same K) is hard. Restrict.

Emergency ArXiv Drop

New work about EIV setting[5,6]

Quadratic Stabilization under a single QMI constraint

Feb 4 and 6, 2024

Based on projection lemma (for $G = G^T \succeq 0$):

$$EE^T \preceq FGF^T \quad \Leftrightarrow \quad \exists D : E = FD,\ DD^T \preceq G \tag{2}$$

[5] L. Li, A. Bisoffi, C. D. Persis, and N. Monshizadeh, "Controller synthesis from noisy-input noisy-output data," 2024.

[6] A. Bisoffi, L. Li, C. D. Persis, and N. Monshizadeh, "Controller synthesis for input-state data with measurement errors," 2024.

Superstablity Definition

System $x_+ = (A + BK)x$ is **superstable**[7,8] if:

- $\|x\|_\infty$ is a Lyapunov Function
- $\|A + BK\|_\infty < 1$

Performance bounds on decay:

$$\|A + BK\|_\infty = \gamma \qquad \Longrightarrow \qquad \|x_t\|_\infty \leq \gamma^t \|x_0\|_\infty$$

[7] F. Blanchini and M. Sznaier. Persistent disturbance rejection via static-state feedback. IEEE Transactions on Automatic Control

[8] B. T. Polyak and P. S. Shcherbakov. Superstable Linear Control Systems. I. Analysis. Automation and Remote Control, 63(8):1239–1254, 2002

Superstability for EIV

Control Problem:

Find $K \in \mathbb{R}^{m \times n}$ s.t. $(A + BK)$ is Superstable $\forall (A, B) \in \mathcal{P}$

Superstability Application

LP with $2n^2 + n$ inequalities over $(A, B, \Delta x, \Delta u) \in \bar{\mathcal{P}}$

$$\begin{aligned}
\operatorname{find}_{K,M} \quad & K \in \mathbb{R}^{m\times n}, \quad M(A,B) : \mathcal{P} \to \mathbb{R}^{n\times n} && (3a)\\
& \forall i = 1..n,\ j = 1..n : \\
& \quad M_{ij}(A,B) \pm (A+BK)_{ij} \geq 0 && (3b)\\
& \forall i = 1..n : 1 - \textstyle\sum_{j=1}^{n} M_{ij}(A,B) > 0 && (3c)
\end{aligned}$$

Nonconservative if $\bar{\mathcal{P}}$ compact (M continuous)

How do we solve infinite LPs?

Discretization necessary to solve on computer

More complexity: more accurate solutions

Method	Increasing Complexity
Gridding	# Grid Points
Basis Functions	# Functions
Random Sampling	# Samples
Sum-of-Squares (SOS)	Polynomial Degree
Your Favorite Method	Some Accuracy Parameter

Runtime usually exponential in dimension, complexity

Computational Complexity (SOS, Full)

Restrict $M_{ij}(A, B)$ to a polynomial of degree $2d$

Each infinite ≥ 0 becomes SOS in $(A, B, \Delta x, \Delta u)$

Required PSD matrix size: $\binom{[n(n+T)+(n+T-1)m]+d}{d}$

Example

$(n = 2, m = 2, T = 15, d = 2)$: size 2278

Too big for interior point methods

Parameterized Robust Counterparts

Motivation and Size Comparison

Maximal size of PSD matrices (SOS)

Size	Full	Robustified
Super	$\binom{[n(n+T)+m(n+T-1)]+d}{d}$	$\binom{[n(n+m)]+d}{d}$

When $(n = 2, m = 2, T = 15, d = 2)$:

$$\text{Full} = 2278, \text{ Robust} = 45$$

Robustified PSD size indepedent of T

Polytopic Consistency Structure

Consistency set $\bar{\mathcal{P}}$:

$$\bar{\mathcal{P}} : \begin{Bmatrix} 0 = -\Delta x_{t+1} + A\Delta x_t + B\Delta u_t + h_t^0 & \forall t = 1..T-1 \\ \|\Delta x_t\|_\infty \leq \epsilon_x, \ \|\Delta u_t\|_\infty \leq \epsilon_u & \forall t = 1..T \end{Bmatrix}$$

$\bar{\mathcal{P}}$ is a $(\Delta x, \Delta u)$-polytope parameterized by (A, B)

Can therefore eliminate $\Delta x, \Delta u$ (and also η if present)

Robust Counterpart Example: Box

Linear inequality constraint in β with robust uncertainty

Original β-feasible problem with unknown $\|w\|_\infty \leq 1$

$$\forall w : \quad a_0^T \beta + \sum_{\ell=1}^{L} w_\ell a_\ell^T \beta \leq b_0 + \sum_{\ell=1}^{L} w_\ell b_\ell \tag{4}$$

Robust Counterpart Example: Box

Original β-feasible problem with unknown $\|w\|_\infty \leq 1$

$$\forall w : \quad a_0^T\beta + \sum_{\ell=1}^{L} w_\ell a_\ell^T\beta \leq b_0 + \sum_{\ell=1}^{L} w_\ell b_\ell \tag{4}$$

Equivalent **Robust Counterpart** with w eliminated

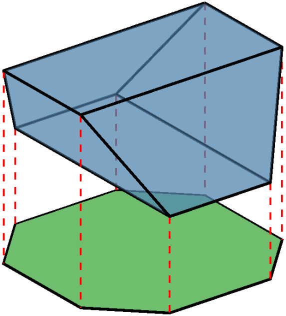
$$\max_{\|w\|_\infty \leq 1} \left(\sum_{\ell=1}^{L} w_\ell [a_\ell^T\beta - b]\right) \leq b_0 - a_0^T\beta \tag{5a}$$

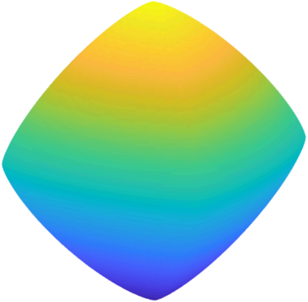
$$\sum_{\ell=1}^{L} |a_\ell^T\beta - b| \leq b_0 - a_0^T\beta \tag{5b}$$

Semidefinite-Representable (SDR) Set

Uncertainty $W \ni w$ (SDR with $K \subseteq \mathbb{S}^{+}$) [9]

$$W = \{w \in \mathbb{R}^{L} \mid \exists \lambda \in \mathbb{R}^{\Lambda} : Cw + G\lambda + e \geq_K 0\} \qquad (6)$$





[9] Fawzi, H., Gouveia, J., Parrilo, P. A., Saunderson, J., Thomas, R. R. (2022). Lifting for simplicity: Concise descriptions of convex sets. SIAM Review, 64(4), 866-918.

Robust Counterparts (General)

Original problem with SDR uncertainty

$$\forall w \in W: \qquad a_0^T \beta + \sum_{\ell=1}^{L} w_\ell a_\ell^T \beta \leq b_0 + \sum_{\ell=1}^{L} w_\ell b_\ell \tag{7}$$

Robust counterpart with dual variable ζ (sufficient)

$$e^T \zeta + a_0^T \beta \leq b_0 \tag{8a}$$

$$G\zeta = 0, \ \zeta \geq_{K^*} 0 \tag{8b}$$

$$(C^T \zeta)_\ell + a_\ell^T \beta = b_\ell \qquad \forall \ell = 1..L \tag{8c}$$

Also necessary if (K convex, pointed, non-polyhedral Slater)[10]

[10] A. Ben-Tal, L. El Ghaoui, and A. Nemirovski, Robust Optimization. Princeton University Press, 2009, vol. 28

Robust Counterparts (Parameterized)

$(C, G, e, a_0, a_\ell, b_0, b_\ell)$ now depends on parameter $p \in P$

Strict robust inequality (with $\zeta(p)$ parameter-dependent):

$$\forall w \in W(p): \; a_0^T(p)\beta + \sum_{\ell=1}^{L} w_\ell a_\ell(p)^T \beta < b_0(p) + \sum_{\ell=1}^{L} w_\ell b_\ell(p)$$

Can we parameterize $\zeta(p)$ by continuous functions (e.g., NN, Polynomial, RBF)?

Conditions for Exactness

Necc. + Suff. conditions for exactness [11] [12]:

1. K convex, pointed, Slater if non-polyhedral
2. P is compact
3. All data continuous in p
4. p-Slater: $\exists \zeta(p) >_{K^*} 0$ with $\forall p : [C(p)^T; G(p)^T]\zeta(p) = 0$

[11] Miller, Jared, and Mario Sznaier. "Analysis and Control of Input-Affine Dynamical Systems using Infinite-Dimensional Robust Counterparts," arXiv:2112.14838, 2023.

[12] Conditions based on: J. Denel, "Extensions of the continuity of point-to-set maps: applications to fixed point algorithms," Point-to-Set Maps and Mathematical Programming, pp. 48–68, 1979

Back to EIV

Robustified Superstabilizing EIV

Each of the $2n^2 + n$ constraints are robustified

Example: $M_{ij} - (A + BK)_{ij} \geq 0 \; \forall (A, B, \Delta x, \Delta u) \in \bar{\mathcal{P}}$

EIV SOS: each per-constraint $\zeta(A, B)$ is a polynomial vector

Process noise alone: ζ is constant in (A, B) [13]

[13] Y. Cheng, M. Sznaier, C. Lagoa, Robust superstabilizing controller design from open-loop experimental input/output data. In IFAC SYSID, 2015.

Other EIV Problems

Robust linear inequalities

- ARX models [14]
- Extended Superstabilization
- Positive Stabilization

Robust LMIs (robust counterparts could be conservative)

- Quadratic stabilization[15]
- Worst-case H_2-optimal control

[14] Jared Miller, Tianyu Dai, and Mario Sznaier. Superstabilizing Control of Discrete-Time ARX Models under Error in Variables. In 22nd IFAC World Congress.

[15] –. Data-Driven Stabilizing and Robust Control of Discrete-Time Linear Systems with Error in Variables, 2023, arXiv:2210.13430.

Examples

Example 1: (Monte Carlo, Superstabilization)

Ground truth system $(\epsilon_\eta, \epsilon_u = 0)$

$$A = \begin{bmatrix} 0.6863 & 0.3968 \\ 0.3456 & 1.0388 \end{bmatrix}, \quad B = \begin{bmatrix} 0.4170 & 0.0001 \\ 0.7203 & 0.3023 \end{bmatrix}$$

$S =$ percentage of success in 50 trials

S vs. ϵ_x with $T = 8$

ϵ_x	0.05	0.08	0.11	0.14
S	100	84	57	39

S vs. T with $\epsilon_x = 0.14$

T	8	10	12	14
S	39	60	75	86

Example 2: Infeasibility of Sole Process Noise

Previous system with

$\epsilon_x = 0.0851,\ \epsilon_u = 0.0489,\ \epsilon_\eta = 0.0192,\ \ T = 8$

EIV superstabilization: $\gamma_{\text{EIV}} = 0.9592$

Process noise describing ground truth (not known a-priori):

$\epsilon_{\text{true}} = 0.2407,\ \gamma_{\text{true}} = 0.8910$

Pure process noise $\supseteq$ EIV: $\epsilon_{\text{wrap}} = 0.3582,\ \gamma_{\text{wrap}} = \infty$

Optimal Control Problems (and Friends)

Optimal Control Setup

Standard optimal control problem (Bolza-form)

$$
\begin{aligned}
P^* = \inf_{u(t)} \int_{t=0}^{T} J(t, x(t), u(t)) dt + J_T(x(T)) \\
\dot{x}(t) = f(t, x(t), u(t)),\ \ u(t) \in U \qquad\qquad \forall t \in [0, T] \\
x(0) = x_0,\ \ x(T) \in X_T
\end{aligned}
$$

Assume input-affine $f(t, x, u) = f_0(t, x) + \sum_{j=1}^{m} u_j f_j(t, x)$

Optimal Control Program

HJB Equations for Value Function V^*

$$
\begin{aligned}
0 &= J_T(x(T)) - V^*(T, x(T)) \\
0 &= \inf_{u \in U} \left(\dot{V}^*(t, x(t), u) + J(t, x(t), u(t)) \right) \quad \forall t \in [0, T]
\end{aligned}
$$

HJB Inequalities for subvalue $v(t, x) \leq V^*(t, x)$

$$
\begin{aligned}
J_T(x) - v(T, x) &\geq 0 \quad \forall x \in X_T \\
J(t, x, u) + \dot{v}(t, x, u) &\geq 0 \quad \forall (t, x, u) \in [0, T] \times X \times U
\end{aligned}
$$

Optimal Control Relaxation

Tightest subvalue at initial condition x_0

$$
\begin{aligned}
p^*(x_0) = \sup_v \; & v(0, x_0) \\
& J_T(x) - v(T, x) \geq 0 && \forall x \in X_T \\
& J(t, x, u) + \dot{v}(t, x, u) \geq 0 && \forall (t, x, u) \in [0, T] \times X \times U \\
& v \in C^1([0, T] \times X)
\end{aligned}
$$

$p^*(x_0) = V^*(0, x_0)$ if (compact, regularity, convexity) [16]

[16] Lewis, Richard M., and Richard B. Vinter. "Relaxation of optimal control problems to equivalent convex programs." Journal of Mathematical Analysis and Applications 74.2 (1980): 475-493.

Robustified Lie Constraint

Assuming that U, $\mathrm{Epigraph}(J)$ are (t, x)-SDR, define

$$\bar{U}(t,x) = \{u \in U(t,x),\ \tau \in [J(t,x,u), \max_{t,x,u}(J)]\} \qquad (9)$$

Epigraph-expanded Lie constraint (includes LQR):

$$\begin{aligned} &\forall (t,x) \in [0,T] \times X,\ \forall (\tau, u) \in \bar{U}(t,x) : \\ &\qquad \tau + (\partial_t + f_0 \cdot \nabla)v + \textstyle\sum_{j=1}^m u_j f_j \cdot \nabla_x v \geq 0 \end{aligned} \qquad (10)$$

Robustify to eliminate (u, τ), $(1 + n)$ variables left (t, x)

Prior Robustified Dynamics

Parameterized and robustified dynamical systems problems

- L_1-optimal OCP with $U = \text{Box}$[17] [18]
- Density Functions with $U = \text{Polytope}$ ($X = \mathbb{R}^n$: Suff.) [19]

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[17] A. Majumdar, R. Vasudevan, M. M. Tobenkin, and R. Tedrake, "Convex Optimization of Nonlinear Feedback Controllers via Occupation Measures," The International Journal of Robotics Research, 2014

[18] M. Korda, D. Henrion, and C. N. Jones, "Controller design and value function approximation for nonlinear dynamical systems", Automatica, 2016

[19] Dai, Tianyu, and Mario Sznaier. "A semi-algebraic optimization approach to data-driven control of continuous-time nonlinear systems." IEEE L-CSS, 2020.

OCP Friends

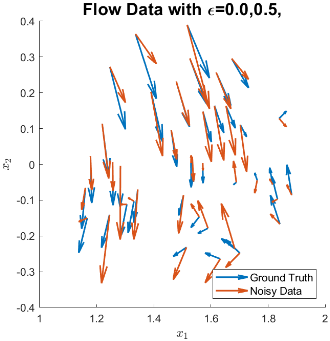
Related problems with decomposable Lie constraints:

- Peak and Distance estimation
- Reachable set estimation
- Maximum controlled invariant set
- Discounted infinite-time optimal control
- Time-delay analysis and control

Data-Driven Analysis

Sampling: Flow System

Data $\mathcal{D} = \{(t_j, x_j, \dot{x}_j)\}_j$ under mixed L_∞-bounded noise



$$\dot{x} = [x_2, -x_1 - x_2 + x_1^3/3]$$

Dynamics Model

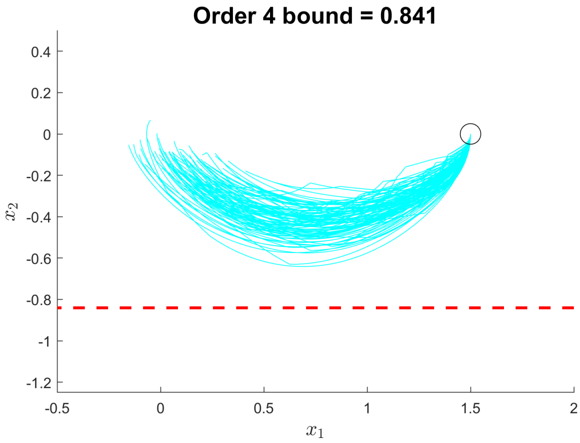
Given data $\mathcal{D}$, budget ϵ, system model $\{f_0, f_\ell\}$

Parameterize ground truth F by functions in dictionary

$$\dot{x}(t) = f(t, x, w) = f_0(t, x) + \sum_{\ell=1}^{L} w_\ell f_\ell(t, x)$$

Treat unknown parameters w as adverse inputs

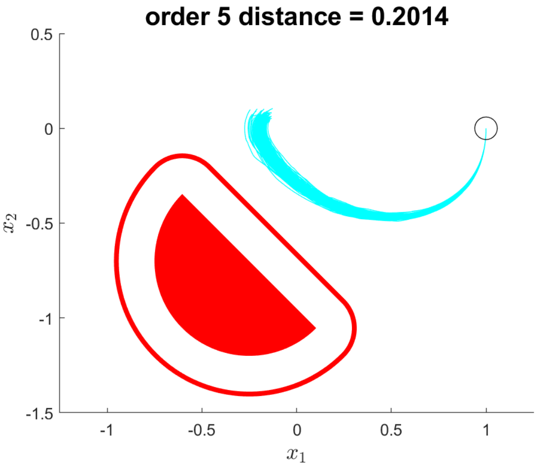
Peak Estimation Example (Flow)



$$\dot{x} = [x_2,\ \text{cubic}(x_1, x_2)]$$

$$L = 10,\ \ m = 80\ (33\ \text{nonredundant})$$

Distance Estimation Example (Flow)

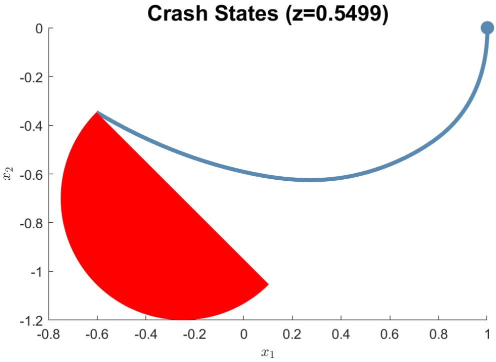


$$\dot{x} = [x_2,\ \text{cubic}(x_1, x_2)]$$

$$L = 10,\ \ m = 80\ (33\ \text{nonredundant})$$

Crash-Bound Estimation (Flow)

Minimum data corruption needed to crash into unsafe set[20]

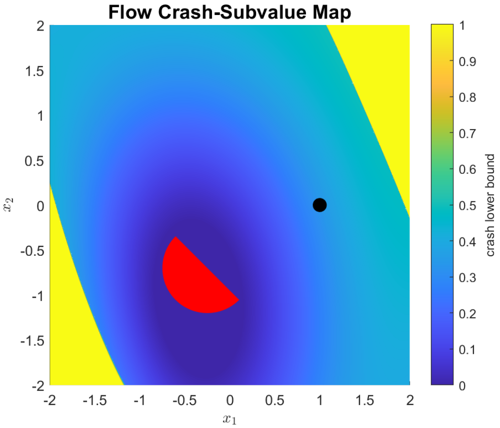


True $\epsilon = 0.5$, distance ≈ 0.2014

[20] Jared Miller and Mario Sznaier. "Quantifying the Safety of Trajectories using Peak-Minimizing Control", arXiv:2303.11896, 2023.

Crash-Subvalue (Flow)

Subvalue map: lower-bound on ϵ needed to crash



Bound at • of $0.3399 \leq 0.5499$, but valid everywhere in X

Discrete-Time OCP/Peak (if time allows)

Lack of Linearity

Discrete-time dynamics $x_+ = f(x,u) = f_0(x) + \sum_j u_j f_j(x)$

If $v(x)$ nonlinear, then $v(f(x,u))$ nonlinear in u

$$\forall (x,u) \in X \times U$$
$$J(x,u) + \Big(v(x) - v(f_0(x,u) + \textstyle\sum_{j=1}^m u_j f_j(x,u))\Big) \geq 0.$$

Obstacle to discrete-time optimal control [21]

[21] W. Han and R. Tedrake, "Controller Synthesis for Discrete-Time Polynomial Systems via Occupation Measures," IEEE IROS, 2018

Linear with expanded state

New variable $\tilde{x}$ with support:

$$\Omega = \{(x, \tilde{x}, u) \in X^2 \times U \mid \tilde{x} = f(x, u)\}$$

Equivalent HJB inequality:

$$J(x, u) + v(x) - v(\tilde{x}) \geq 0 \qquad \forall (x, \tilde{x}, w) \in \Omega.$$

Robustify to eliminate u, left with $(t, x, \tilde{x})$

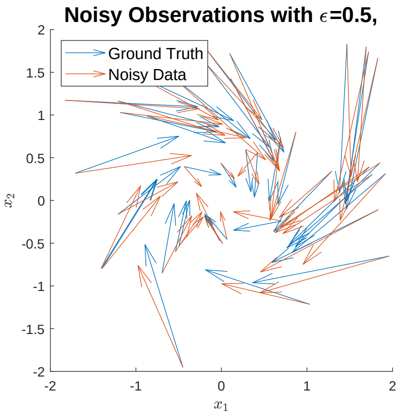
Unlocked Possibilities

Favorable when $m > n$, substitute $(t, x, u) \rightarrow (t, x, \tilde{x})$

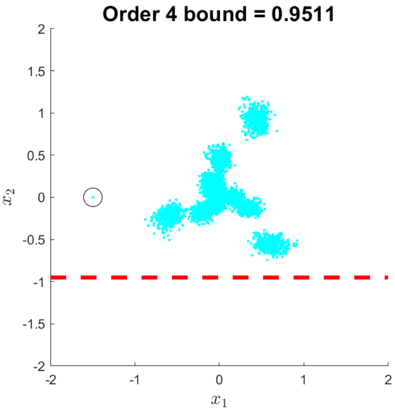
Optimal control and data-driven analysis of:

- Discrete-time systems
- Hybrid systems (resets)
- Stochastic analogues (jump maps)

Peak Estimation in Discrete-Time



(a) Observed data of



(b) Order 4 tightening

$$\text{Quadratic Model of } x_+ = \begin{bmatrix} -0.3x_1 + 0.8x_2 + 0.1x_1x_2 \\ -0.75x_1 - 0.3x_2 \end{bmatrix}$$

Take-aways

Conclusion

Stabilization in the Error-in-variables setting

Robustification to reduce complexity

Can apply robustification to many other problems

Thank you for your attention

Questions?

Bonus (Sum-of-Squares)

Sum-of-Squares Method

Every $c \in \mathbb{R}$ satisfies $c^2 \geq 0$

Sufficient: $q(x) \in \mathbb{R}[x]$ nonnegative if $q(x) = \sum_i q_i^2(x)$

Exists $v(x) \in \mathbb{R}[x]^s$, *Gram* matrix $Z \in \mathbb{S}_+^s$ with $q = v^T Z v$

Sum-of-Squares (SOS) cone $\Sigma[x]$

$$\begin{aligned} & x^2y^4 - 6x^2y^2 + 10x^2 + 2xy^2 + 4xy - 6x + 4y^2 + 1 \\ = & (x+2y)^2 + (3x - 1 - xy^2)^2 \end{aligned}$$

Motzkin Counterexample (nonnegative but not SOS)

$$x^2y^4 + x^4y^2 - x^2y^2 + 1$$

Sum-of-Squares Method (cont.)

Putinar Positivestellensatz (Psatz) nonnegativity certificate over set $\mathbb{K} = \{x \mid g_i(x) \geq 0, h_j(x) = 0\}$ ($\Sigma[\mathbb{K}]$:

$$q(x) = \sigma_0(x) + \textstyle\sum_i \sigma_i(x) g_i(x) + \sum_j \phi_j(x) h_j(x)$$
$$\exists \sigma_0(x) \in \Sigma[x], \quad \sigma_i(x) \in \Sigma[x], \quad \phi_j \in \mathbb{R}[x]$$

Psatz at degree $2d$ is an SDP, monomial basis: $s = \binom{n+d}{d}$

Archimedean: $\exists R \geq 0$ where $R - \|x\|_2^2$ has Psatz over $\mathbb{K}$

$\mathbb{K}$ Archimedean: all $\mathbb{K}$-positive polynomials are in $\Sigma[\mathbb{K}]$

Polynomial Matrix Inequalities

SOS method (scalar): $q(x) \geq 0$

Extend to matrices $Q(x) \in \mathbb{S}^s_{++}$

SOS matrix: $Q(x) = R(x)^T R(x) \in \Sigma^s[x]$ for matrix $R(x)$

Gram matrix (PSD) constraint of size $s\binom{n+d}{d}$

Scherer Psatz: nonnegativity over constraint sets

Bonus (Preprocessing)

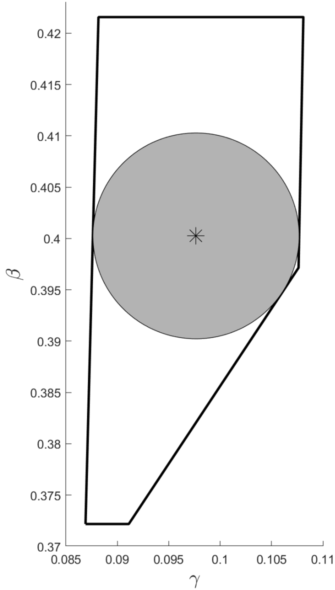
Preprocessing: Centering

Chebyshev center c: center of sphere with largest radius in W

Find through linear programming

$$\max r$$
$$A_k c + r\|A_k\|_2 \leq b_k \qquad \forall k$$
$$r \geq 0, c \in \mathbb{R}^L$$

Shifted dynamics $f_0 \leftarrow f_0 + \sum_{\ell=1}^{L} c_\ell f_\ell$

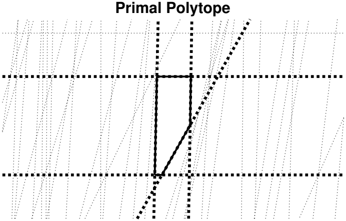


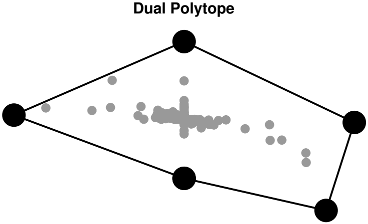
Preprocessing: Redundancy

Majority of $m = 2N_x N_s$ constraints are often redundant

Convex hull of dual polytope:
Time: $\Omega(m \log m + m^{\lfloor L/2 \rfloor})$

Linear program per constraint:
Time: $m \times \tilde{O}(mL + L^3)$ [a]
(Jan van den Brand *et. al.* 2020)





[a] van den Brand, Jan, "A deterministic linear program solver in current matrix multiplication time". Proceedings of the Fourteenth Annual ACM-SIAM Symposium on Discrete Algorithms, 2020.

Bonus (EIV Robustified Program)

EIV Robustified Program

Robustified condition for $q(A, B) \geq 0, \ \forall (A, B, \Delta x, \Delta u) \in \mathcal{P}$.

Find $\zeta^{\pm}_{1:T}(A, B) \geq 0, \ \psi^{\pm}_{1:T-1}(A, B) \geq 0, \ \mu_{1:T-1}(A, B)$

$$
\begin{aligned}
& q \geq \textstyle\sum_{t=1}^{T-1} \mu_t^T h_t^0 + \sum_{t,i} \epsilon_x (\zeta^+_{t,i} + \zeta^-_{t,i}) && \forall (A, B) \\
& \qquad + \textstyle\sum_{t,i} \epsilon_u (\psi^+_{t,i} + \psi^-_{t,i}) && \\
& \zeta^+_1 - \zeta^-_1 = A^T \mu_1 && \\
& \zeta^+_T - \zeta^-_T = -\mu_{T-1} && \\
& \zeta^+_t - \zeta^-_t = A^T \mu_t - \mu_{t-1} && \forall t \in 2..T-1 \\
& \psi^+ - \psi^-_t = A^T \mu_t - \mu_{t-1} && \forall t \in 1..T-1
\end{aligned}
$$

EIV Robustified Program with Process Noise

Includes $\|\eta\|_\infty \leq \epsilon_\eta$, multipliers $\mu^\pm$ with $\Delta\mu = \mu^+ - \mu^-$.

Find $\zeta^\pm_{1:T}(A, B) \geq 0$, $\psi^\pm_{1:T-1}(A, B) \geq 0$, $\mu_{1:T-1}(A, B) \geq 0$

$$q \geq \sum_{t=1}^{T-1} (\mu_t^+ + \mu_t^-)^T h_t^0 + \sum_{t,i} \epsilon_x (\zeta^+_{t,i} + \zeta^-_{t,i}) \quad \forall (A, B)$$
$$+ \sum_{t,i} \epsilon_u (\psi^+_{t,i} + \psi^-_{t,i})$$
$$\zeta_1^+ - \zeta_1^- = A^T \mu_1$$
$$\zeta_T^+ - \zeta_T^- = -\Delta\mu_{T-1}$$
$$\zeta_t^+ - \zeta_t^- = A^T \Delta\mu_t - \Delta\mu_{t-1} \qquad \forall t \in 2..T-1$$
$$\psi^+ - \psi_t^- = A^T \Delta\mu_t - \Delta\mu_{t-1} \qquad \forall t \in 1..T-1$$

Bonus (ARX model)

ARX model

Autoregressive Model with Exogenous Input (ARX)

$$y_t = -\sum_{i=1}^{n_a} a_i y_{t-i} + \sum_{i=1}^{n_b} b_i u_{t-i}$$

Data $\mathcal{D} = (\hat{u}, \hat{y})$ and no state x, and no state x,

$$\begin{aligned} \hat{u} &= u + \Delta u, & \|\Delta u\|_\infty &\leq \epsilon_u \\ \hat{y} &= y + \Delta y, & \|\Delta y\|_\infty &\leq \epsilon_y \end{aligned}$$

Find controller u to stabilize (a, b) consistent with $\mathcal{D}$

Superstability for ARX

Original model with vectors (a, b)

$$y_t = -\sum_{i=1}^{n_a} a_i y_{t-i} + \sum_{i=1}^{n_b} b_i u_{t-i}.$$

Transfer Function with one-step-behind operator $\lambda u_t = u_{t-1}$

$$G(\lambda) = \frac{\sum_{i=1}^{n_b} b_i \lambda^i}{1 + \sum_{i=1}^{n_a} a_i \lambda^i} = \frac{B(\lambda)}{1 + A(\lambda)}$$

Superstability definition, linear constraints

$$\|a\|_1 < 1$$

Dynamic Compensation

Compensator $C(\lambda) = \tilde{B}(\lambda)/(1 + \tilde{A}(\lambda))$

Closed-loop system

$$G_{cl}(\lambda) = \frac{G(\lambda)}{1 + G(\lambda)C(\lambda)} = \frac{B(\lambda)(1 + \tilde{A}(\lambda))}{(1 + A(\lambda))(1 + \tilde{A}(\lambda)) + B(\lambda)\tilde{B}(\lambda)}$$

Superstable: coefficients of G_{cl} denominator have L_1 norm < 1

Fixed C superstabilizes all $(A, B) \in \mathcal{P}$ (from $\mathcal{D}$)

ARX Program Sizes

Set $\mathcal{P}$ originally contains $(a, b, \Delta u, \Delta y)$

Eliminate $(\Delta u, \Delta y)$ in alternatives

Maximal size of Gram (PSD) matrices $(N = N_a + N_b)$

Size	Full	Alternatives
Super	$\binom{2(N+T)-1+d}{d}$	$\binom{N+d}{d}$

No conservatism in Alternatives

Bonus (OCP)

Assumptions

1. U and Graph(J) are (t, x)-SDR
2. $[0, T] \times X \times U$ is compact
3. Input-affine dynamics $f = f_0 + \sum_{j=1}^m u_j f_j$
4. Each f_0, f_ℓ is continuous
5. Original HJB inequality relaxation is tight

Assumption 1: L_1-optimal, L_∞-optimal, LQR, etc.

Bonus (Crash-Safety)

Crash-Safety

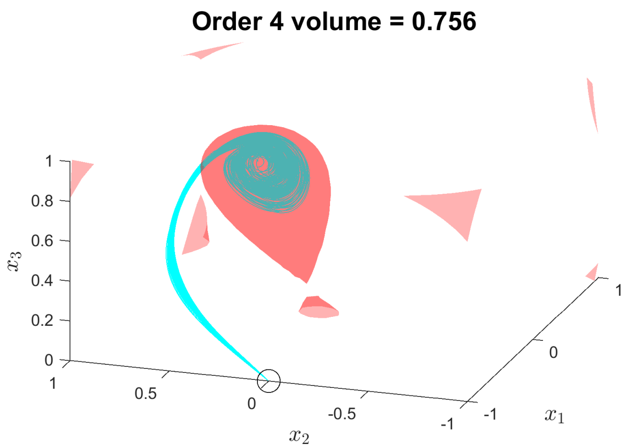
Corruption in L_∞-bounded setting

$$
\begin{aligned}
J(w) &= \max_k \| f_0(t_k, x_k) + \textstyle\sum_{\ell=1}^{L} w_\ell f_\ell(t_k, x_k) - y_k \|_\infty \\
&= \max(h - \Gamma w) \quad \text{for some polytope} \;\; (\Gamma, h)
\end{aligned}
$$

How much data corruption is needed to crash?

$$
\begin{aligned}
Q^* = \inf_{t,\; x_0,\; w} \; & \sup_{t' \in [0,t]} \; J(w(t')) \\
& \dot{x}(t') = f(t', x(t'), w(t')) \qquad\qquad \forall t' \in [0, T] \\
& x(t \mid x_0, w(\cdot)) \in X_u \\
& w(\cdot) \in W, \;\; t \in [0, T], \;\; x_0 \in X_0
\end{aligned}
$$

Reachable Set Estimation Example (Twist)

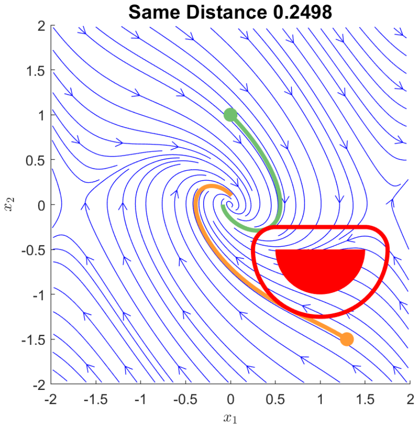


Unknown A, Known B

$L = 9,\ m = 600$ (34 nonredundant)

Example Crash-Bounds

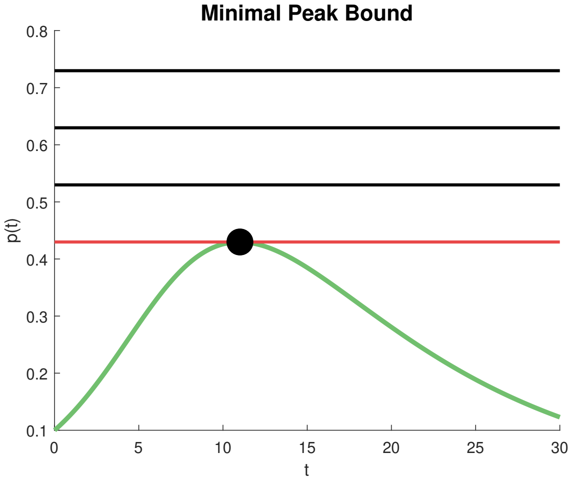
Two trajectories have same distance, different crash-bounds



Green-Top $Q^* = 0.316$, Yellow-Bottom $Q^* = 0.622$

Peak Minimizing Control

Find minimum bound on the maximum p value



Crash-safety is Peak Minimizing Control

Peak-Minimizing Control

Add state $\dot{z} = 0$ (Molina, Rapaport, Ramírez 2022)

$$
\begin{aligned}
Q_z^* = \inf_{t,\ x_0,\ z,\ w} \ & z \\
& \dot{x}(t') = f(t', x(t'), w(t')) & \forall t' \in [0, T] \\
& \dot{z}(t') = 0 & \forall t' \in [0, T] \\
& J(w(t')) \leq z & \forall t' \in [0, T] \\
& x(t \mid x_0, w(\cdot)) \in X_u \\
& w(\cdot) \in W, \ \ t \in [0, T] \\
& x_0 \in X_0, z \in [0, J_{\max}]
\end{aligned}
$$

Drive down the z-upper-bound on $J(w)$

Crash-Bound Program

Consistency sets

$$Z = [0, J_{\max}] \qquad \Omega = \{(w, z) \in W \times Z : J(w) \leq z\}.$$

Optimal Control Problem with auxiliary $v(t, x, z) \in C^1$

$$\begin{aligned} d^* = \sup_{\gamma \in \mathbb{R},\ v} \quad & \gamma \\ & v(0, x, z) \geq \gamma && \forall (x, z) \in X_0 \times Z \\ & v(t, x, z) \leq z && \forall (t, x, z) \in [0, T] \times X_u \times Z \\ & \mathcal{L}_f v(t, x, z, w) \geq 0 && \forall (t, x, z, w) \in [0, T] \times X \times \Omega \end{aligned}$$

Crash Lie-decomposition

Exploit affine structure of $J(w) = \max_j (h - \Gamma w)_j$

Nonconservatively robustified Lie constraint

$$
\begin{aligned}
d^* = \sup_{\gamma \in \mathbb{R},\ v} \quad & \gamma \\
& v(0, x, z) \geq \gamma && \forall (x, z) \in X_0 \times Z \\
& v(t, x, z) \leq z && \forall (t, x, z) \in [0, T] \times X_u \times Z \\
& \mathcal{L}_{f_0} v - (z\mathbf{1} + h)^T \zeta \geq 0 && \forall (t, x, z) \in [0, T] \times X \times [0, J_{\max}] \\
& (\Gamma^T)_\ell \zeta + f_\ell \cdot \nabla_x v = 0 && \forall \ell = 1..L \\
& \zeta_j \in C_+([0, T] \times X \times Z) && \forall j = 1..2nT
\end{aligned}
$$

Bonus (Image Credit)

Drones

designed by freepik.com